

Performance Report

Evaluation

Installation	Equipment	Type	No.
APL DETROIT	MAIN ENGINE	10S90ME-C	DML0105360

Readings Included

	Date	Voyage / title	Remarks
1	2025-11-24	04FLOE1MA	

Main indications (all Readings)

Reading	1
Description	Most recent
Fuel oil saving potential	
Light running	Low
Fuel index ECU	
Firing press. AVG	
Compression press. AVG	
Scav. air press. RECEIVER	
Exh. gas temp. TC in AVG	
Exh. gas temp. TC out AVG	
TC speed AVG	
Spec. FO consumption	

Indications (for Reading 1)

Description	Indication	Comments / recommendations
Light running	Low	Operation point left of nominal propeller curve.
Cool. water temp. CYL out AVG	Very low	Setting of temp. controller low.
ENGINE load	High	please refer to "Light running" in "Main indications"
Engine speed	Low	please refer to "Light running" in "Main indications"

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Optimization potential

Influenced by		Quantification		Proposed actions to achieve the saving / reduction
		Fuel oil equiv.	Thermal load	
Firing pressure by injection		0.0 g/kWh	0 °C	
Firing pressure by compression ratio		0.0 g/kWh	0 °C	
Firing pressure by scavenge air pressure		0.0 g/kWh	0 °C	
ENGINE ROOM ventilation		0.0 g/kWh	0 °C	
Suction press. air TC in				
Press. drop accross SAC				
Scav. air : water temp. SAC in	0.2 g/kWh	1 °C	Reduce SAC water inlet temperature to 34 °C to reduce scav. air temp. by 2 °C.	
Scav. air : temp. RECEIVER	0.1 g/kWh	1 °C	Increase water flow, clean SAC water- and/or air side to reduce scav. air temp. by 1 °C. Target scav. air temp. is 38 °C (incl. reduction of 2 °C by lower SAC cooling water inlet temp. setpoint).*	
Press. drop accross ENGINE	0.0 g/kWh	0 °C		
Exh. gas press. TC out				
Light running	0.0 g/kWh	-		
Fuel oil visc. ENGINE in	0.0 g/kWh	0 °C		
	Fuel oil equiv.	Thermal load	Cyl. lub. oil	
TOTAL (specific)	0.3 g/kWh	-		
TOTAL (of current)	0.2 %	-		CO2 (emission)
TOTAL optimization potential	0.16 t/d	1 °C		SOx (emission)
				0.5 t/d
				2 kg/d
	Cost			
Fuel oil price				
TOTAL saving potential				

Consumption

Measured / calculated values			
Indication	Fuel oil		Cylinder lub oil
Specific consumption reference	169.4 g/kWh		0.80 g/kWh
Specific consumption ISO / MEAS	170.6 g/kWh		0.88 g/kWh
Consumption per hour day *	4440 kg/h	106.6 t/d	21.3 kg/h
Consumption nautical *	247 kg/nmi		
* Measured values, uncorrected			

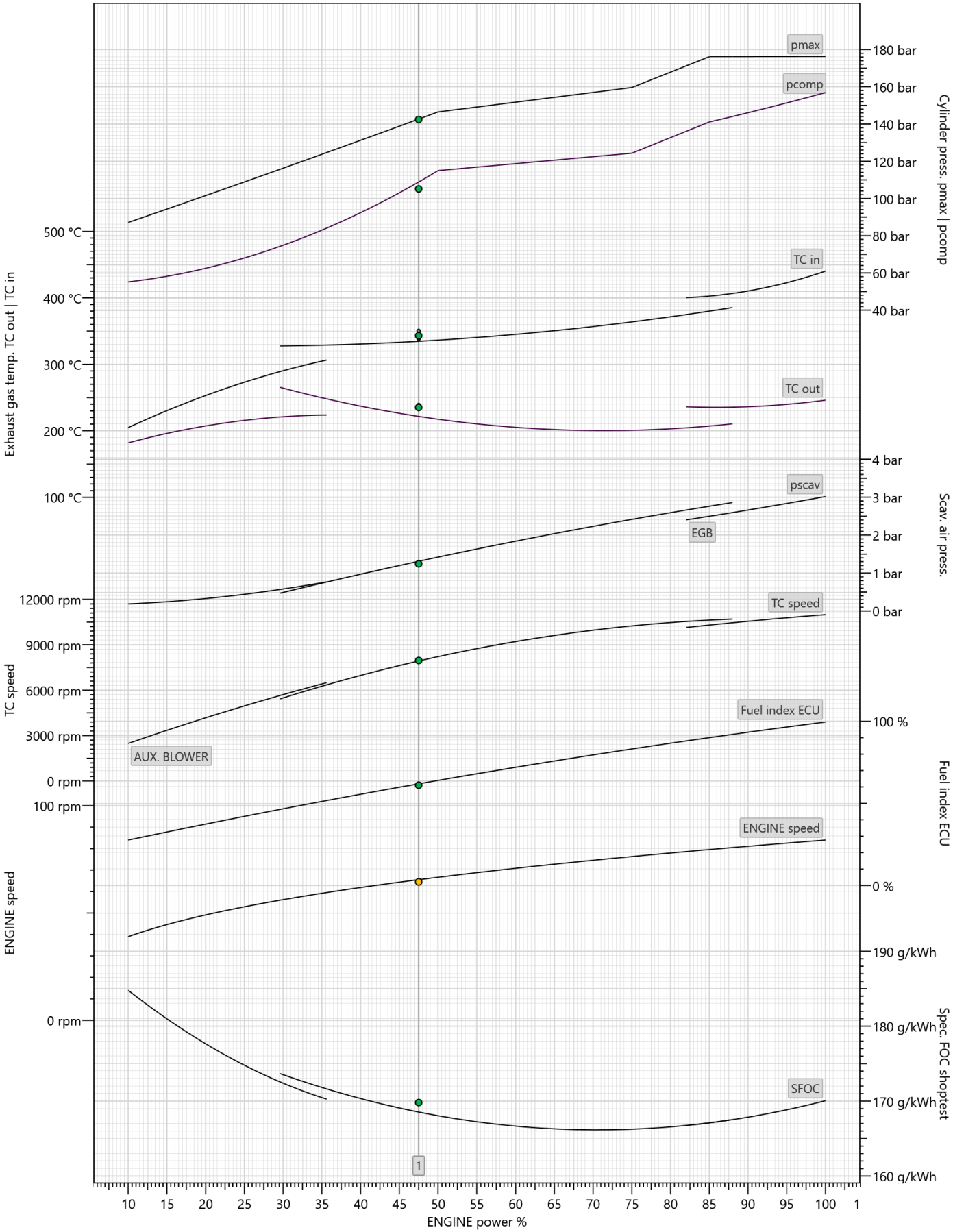
Injection timing and compression ratio adjustment

Cylinders												
Settings		1	2	3	4	5	6	7	8	9	10	
Proposed adjustments to reach target firing press., positive value means increase of pressure:												
Compr. ratio offset [-]		0.0	-0.1	0.1	-0.1	-0.1	0.1	0.0	0.0	0.1	0.1	
pmax offset [bar]		-0.6	0.1	0.3	0.2	-0.5	0.6	0.4	-0.1	-0.3	-0.2	
Common compr. ratio offset [-]	-0.1											
Common pmax offset [bar]	0.5											

Performance Report

Performance Curves

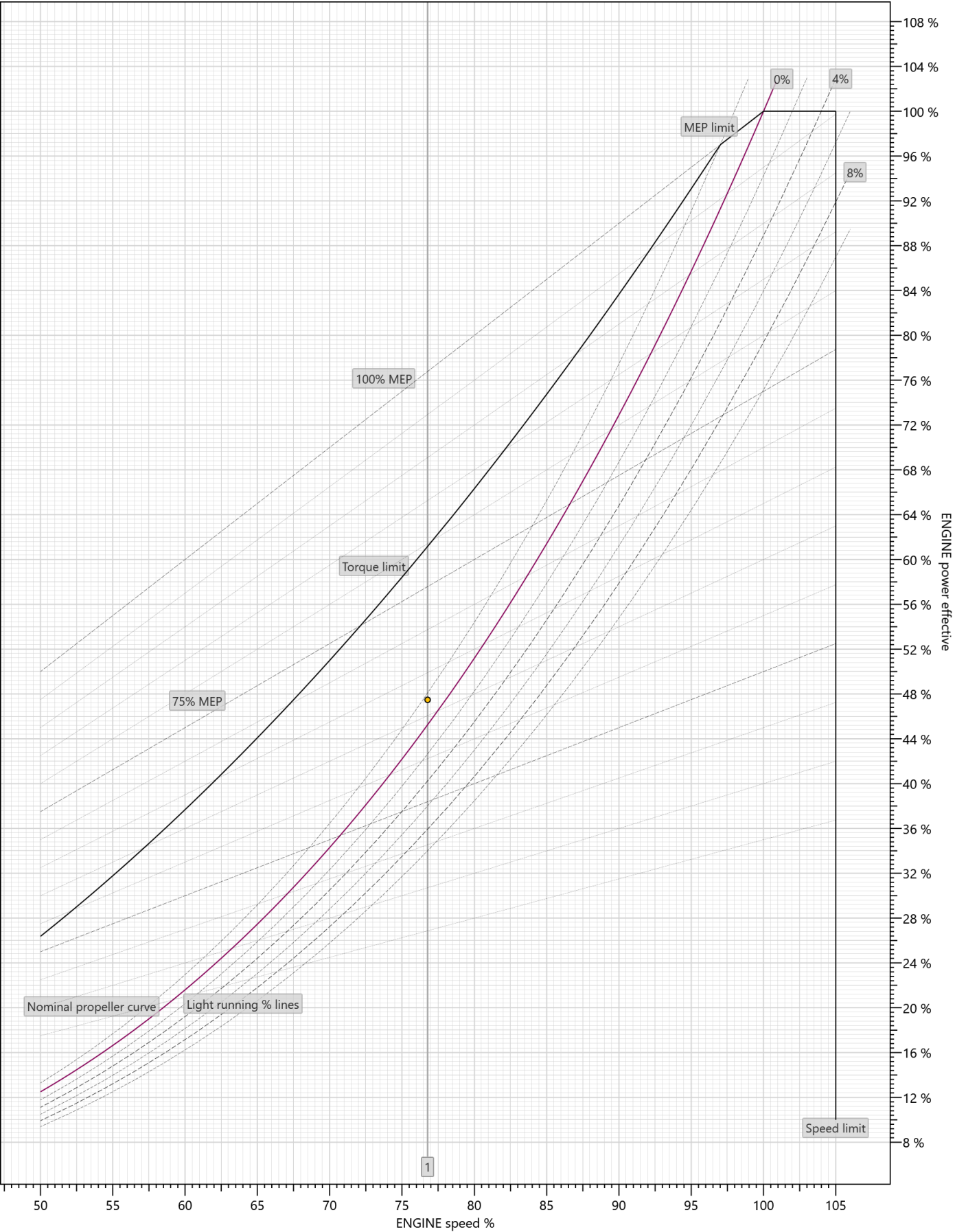
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Load Diagram

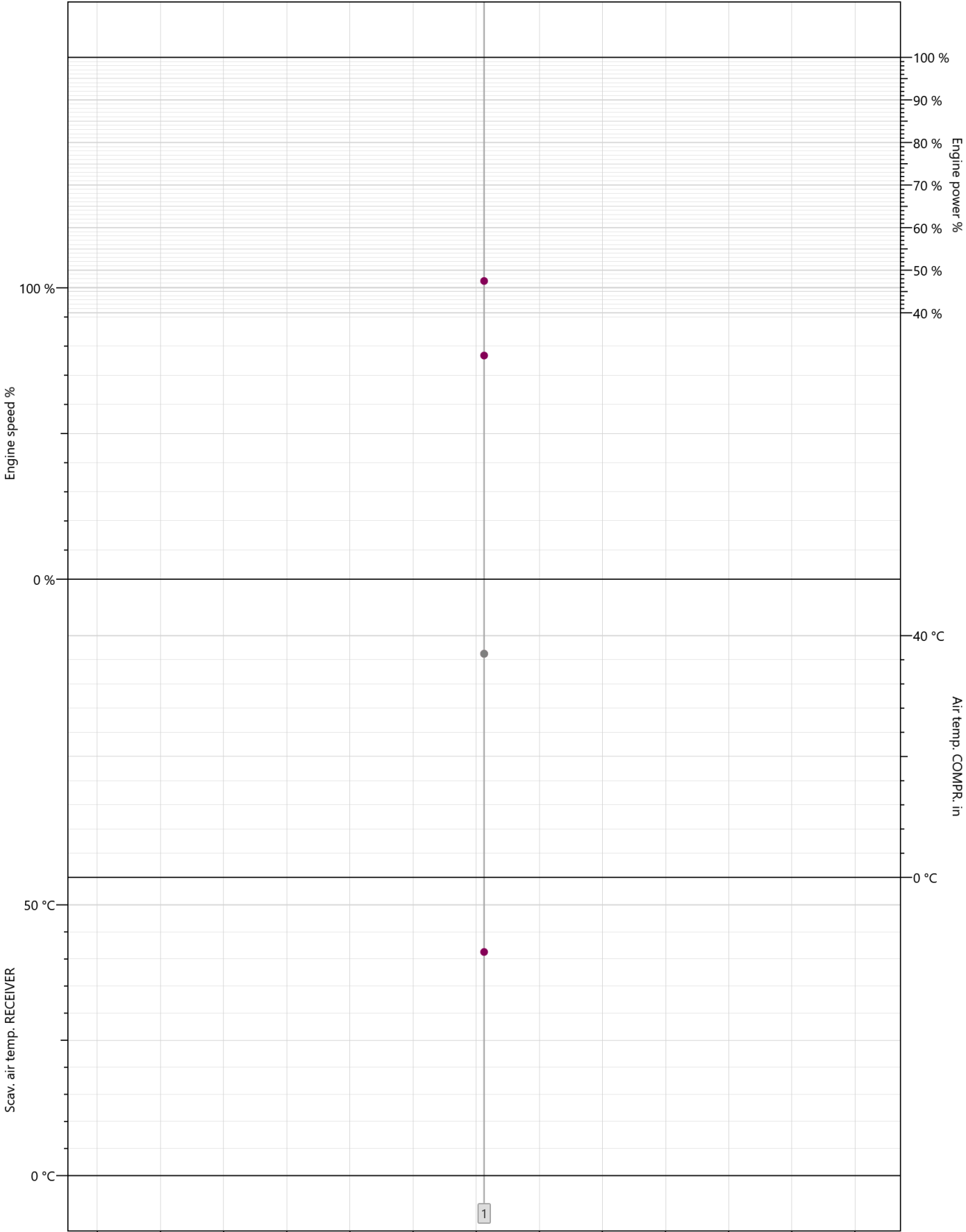
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Trend: LOAD and AMBIENT

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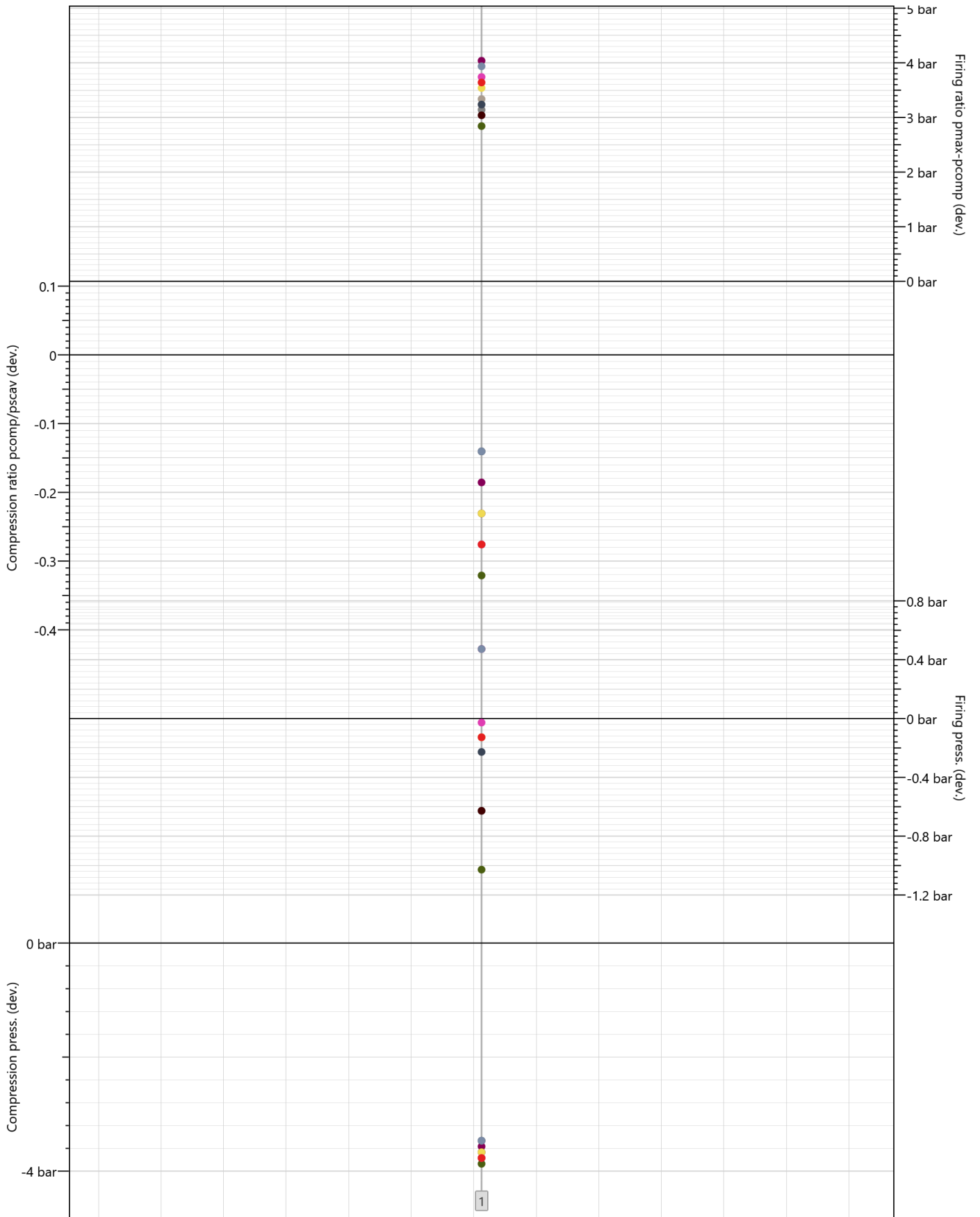
November 2025

TC1	TC2	TC3	AVG	RECEIVER
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Trend: COMBUST. CHAMBER

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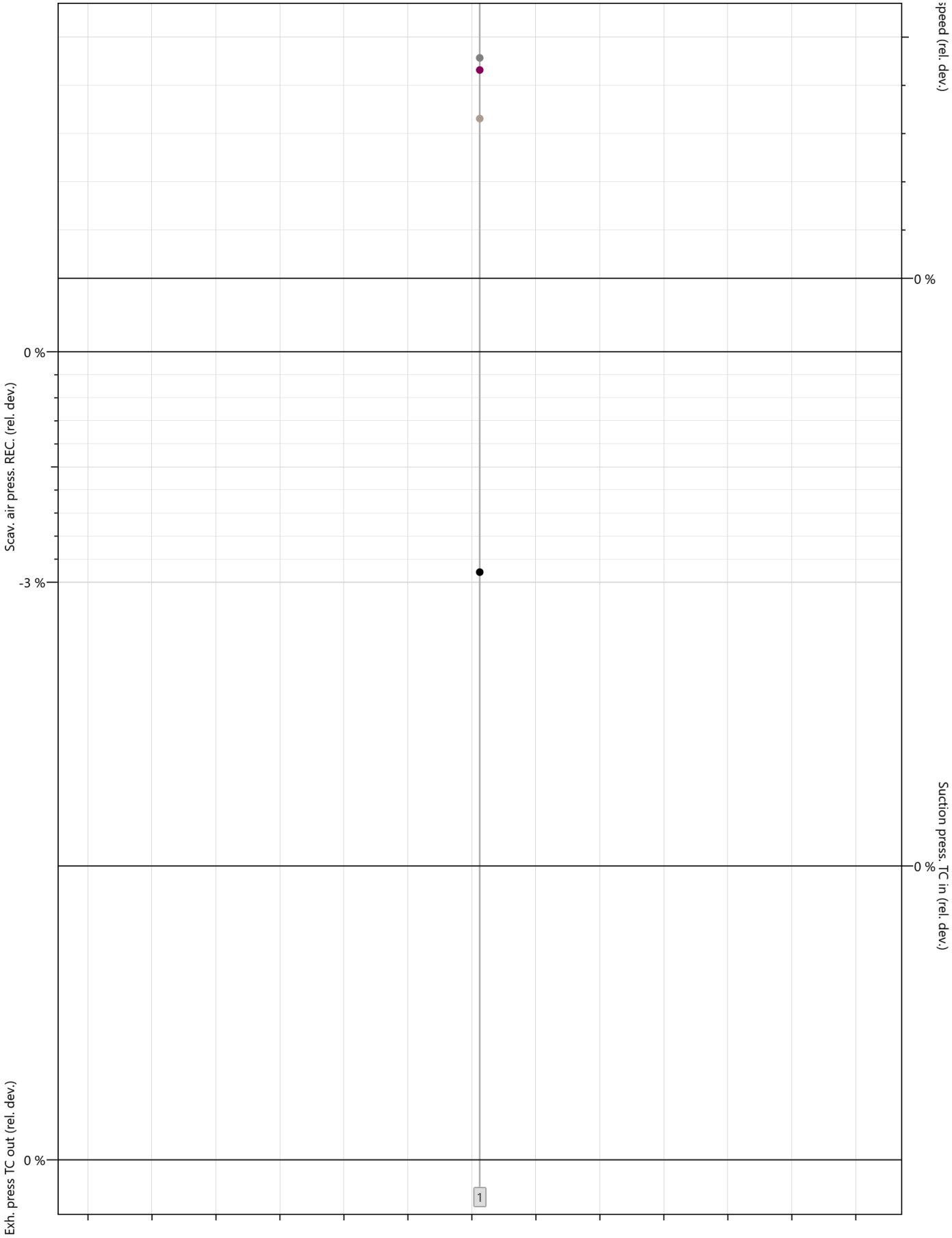
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CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8
CYL 9	CYL 10	AVG (CALC.)					

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Trend: TURBOCHARGER

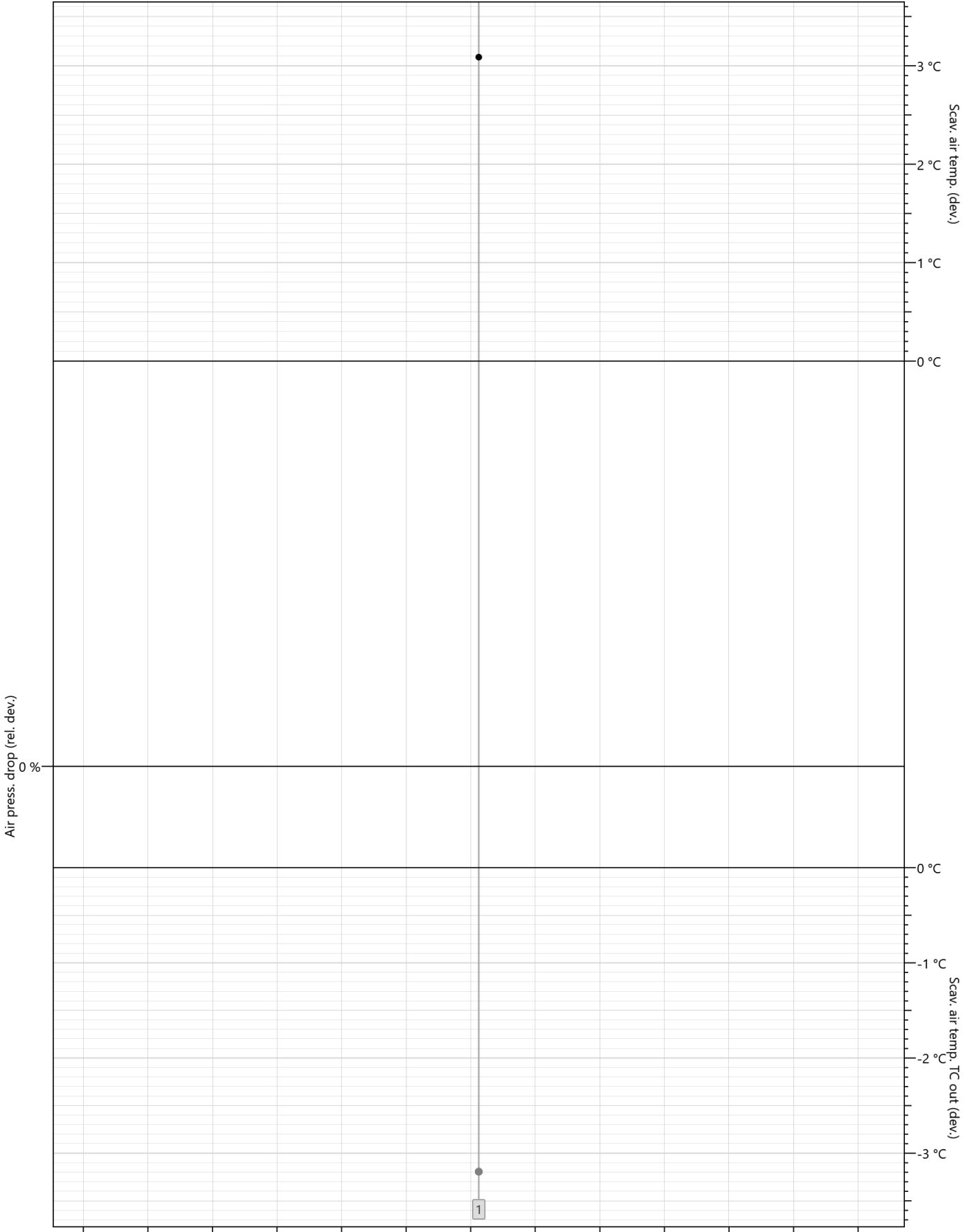
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Trend: SCAV. AIR COOLER

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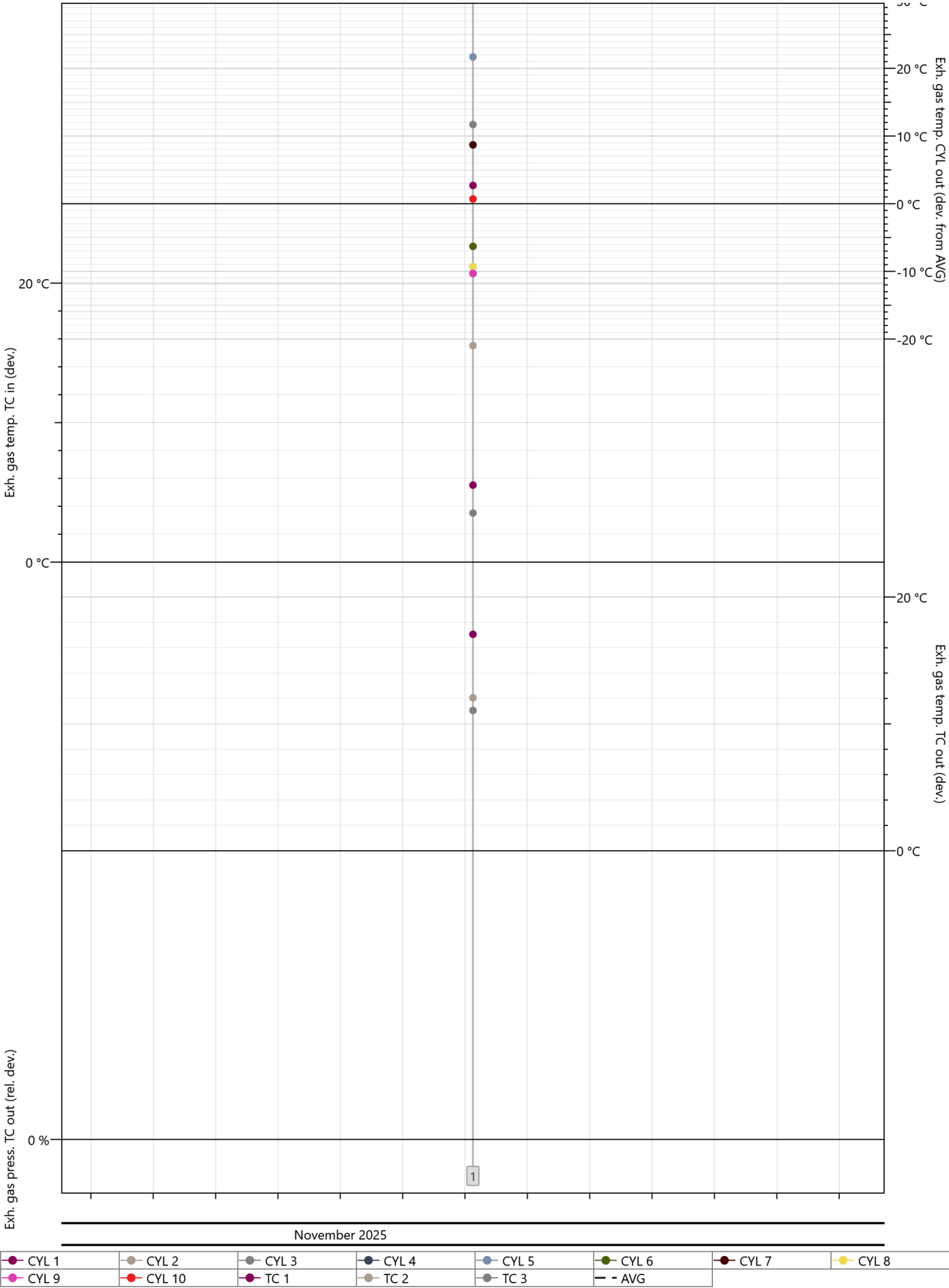
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TC 1	TC 2	TC 3	RECEIVER	- AVG
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Trend: EXHAUST SYSTEM

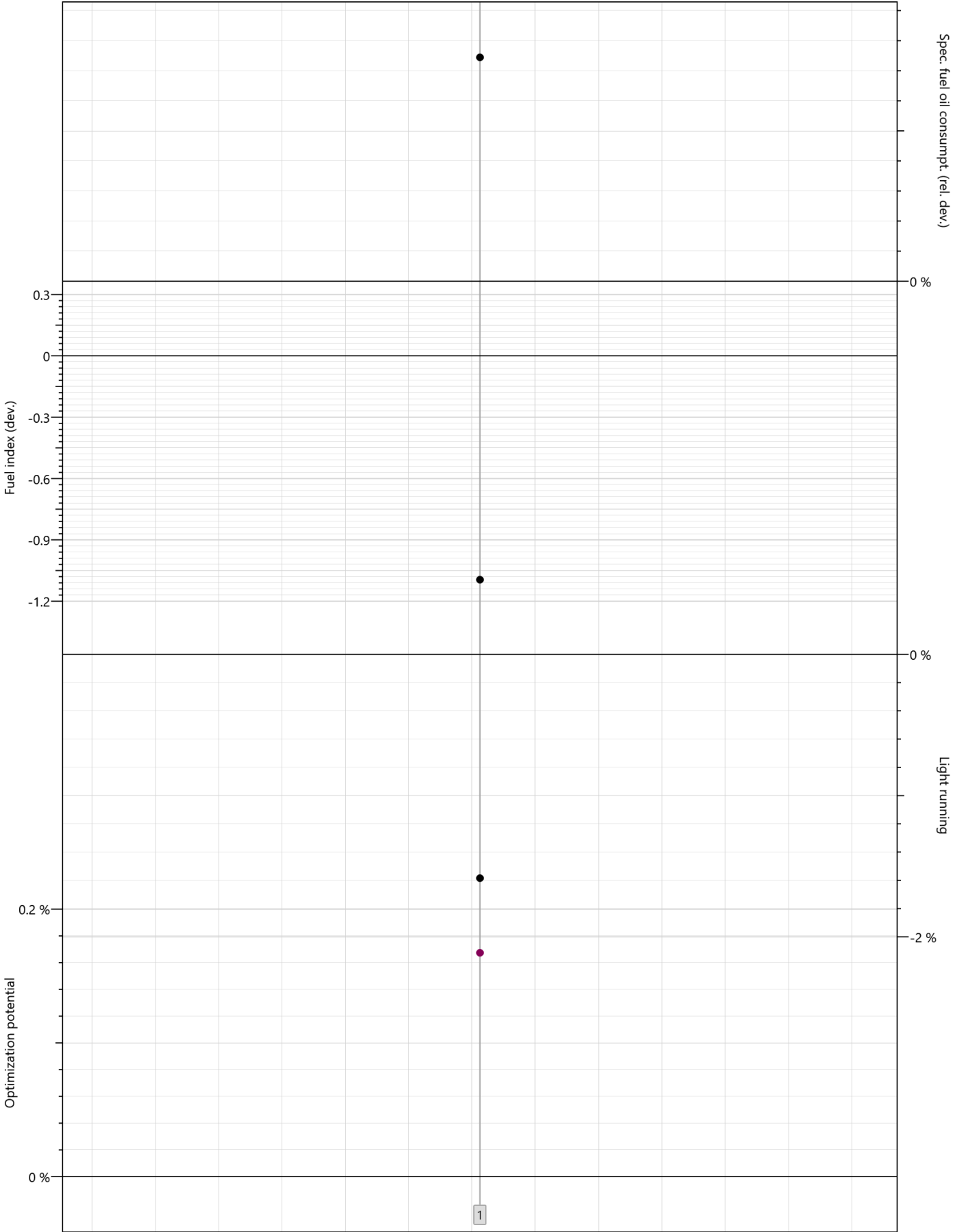
Installation	Equipment	Type	No.
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Trend: CONSUMPTION

Installation	Equipment	Type	No.
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November 2025

Performance Report

Reading

Installation			Equipment			Type			No.																					
APL DETROIT			MAIN ENGINE			10S90ME-C			DML0105360																					
General	Date and time of recording		11/24/2025 15 : 08 : 30		Sea water temp.		28.0 °C		SHIP SOG STW		18.50 18.00 kn		Voy. / title		04FLOE1MA															
	ENGINE state		stable		Ambient air temp.		30.0 °C		Draft fwd aft		10.70 10.90 m		Record by		2E															
	ENGINE running hours		71503 h		Engine room air temp.		37.5 °C		Draft mid Trim		10.80 0.20 m		Ackn. by																	
	Ambient humidity		rel %		Barom. pr. Eng. room		1.016 1.022 barA		Remarks																					
		abs				g/kg																								
Power / Speed			CALCULATED		MEAS		ESTIMATED 24.3(E264X126)																							
			XPERT		TORSIOM.		MOP		by FI*rpm		by pscav		by TC rpm		by MEP		by rpm													
	ENGINE power estimated		kW		24245		23970		24514		23768		23445		24207															
			%		47.5				48.0																					
			MCR		ENGINE		TORSIOMETER		MARGINS		MECH. EFFICIENCY																			
ENGINE speed		rpm		84.0 76.8 %		64.5		Power kW		23970		Light running		-1.6 %		Theoretic.														
ENGINE power effective		kW		51070 47.5 %		24245						Power		-4.9 %																
(for PTI operation, use negative value, "-")																														
Electronic ctrl.	Running mode		Economy		ISO CORRECTED		MEASURED																							
	Governor mode		RPM		REF.		CALC.		ALL																					
	Fuel index ECU		%		62.1		61.1		61.0																					
	Applied fuel quality offset		%						0.0																					
Cylinder pressure	SOURCE:		ISO CORRECTED		MEASURED												SETPOINT													
	QUALITY Manually		REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10		MOP	
	Firing press. pmax		barG		142.6		142.4		142.4		143.1		142.5		142.0		142.4		143.1		141.6		142.0		142.5		142.6		142.5	
	pmax deviation		bar				ALL		0.7		0.1		-0.4		0.0		0.7		-0.8		-0.4		0.1		0.2		0.1			
	pmax offset		bar				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Compression press.		barG		108.8		105.2		103.8		103.9		104.0		103.7		104.0		104.0		103.6		103.8		103.8		103.7		103.7	
	pcomp deviation		bar				ALL		0.1		0.2		-0.1		0.2		0.2		-0.2		0.0		0.0		-0.1		-0.1			
	pcomp / pscav offset		-				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Mean indicated press.		barG		11.56																									
	MIP deviation		bar																											
	Fuel offset high		%				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Fuel offset low		%				MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Power indicated		kW				SUM:																							
	Mean effective press. (MEP)		bar		10.88																									
	pmax-pcomp pcomp / pscav		-		33.8 47.5				38.6 47.3																		pcomp / pscav			
Scavenge air	AUX. BLOWER operation		off		REF.		CALC.		AVG.		TC 1		TC 2		TC 3															
	Suction press.		mmWG		23																									
	Press. drop accr. SAC		mmWG		142																									
	Scav. air press. RECEIVER		barG		1.31		1.24		1.20																					
	Scav.air press.setp. MOP		barG																											
	Air temp. COMPR. in		°C						37.0		37.0		37.0		37.0															
	Scav. air temp. COMPR. out		°C		129				126		126		126		126															
	Scav. air temp. SAC out		°C																											
	Scav. air temp. RECEIVER		°C		40.2				41.3																					
	Cool. water temp. SAC in		°C		34				35.7		35.9		35.5		35.8															
	Cool. water temp. SAC out		°C						40.4		40.7		40.0		40.4															
	Exhaust gas			ISO CORRECTED		MEASURED																								
		REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10				
Exh. gas temp. CYL out		°C		285		285		307		310		298		319		297		329		301		316		298		297		308		
temp. deviation		°C				ALL		3		-9		12		-10		22		-6		9		-9		-10		1				
Exh. vlv. open offset		°CA				MOP SETTING:																								
Exh. gas press. MANIFOLD		barG		1.12		1.07		1.03		TC 1		TC 2		TC 3																
Exh. gas press. TC out		mmWG		68																										
Exh. gas temp. TC in		°C		335		343		366		363		373		361																
Exh. gas temp. TC out		°C		222		235		257		261		256		255																
TC speed		rpm		7928		7960		7977		7979		7971		7981																
Turbocharging efficiency		%		68.6																										
Exh.wastegate (EGB) pos. area %		%		0 0 0 0				0																						
Fuel oil				ISO CORRECTED		MEAS		Fuel oil properties:		ANALYS. ESTIMATE		Fuel oil system:		REF		MEAS		EXPECT.												
	Absolute consumption		m³/h		REF.		CALC.		4.750		Viscosity		cSt		12.5		12.0		17.1											
	Specific FOC		g/kWh		169.4		170.6		183.1		Sulfur content		%		0.49 0.49		Temp. ENGINE in		°C		133		120							
	Light running correction		g/kWh		0.0						Water content		%		0.30 0.30		Temp. FLOWMETER		°C				110							
	Thrust bearing loss		g/kWh		-0.8						LCV		kJ/kg		40423		Density FLOWM.		kg/m³				934.8							
	Specific FOC shoptest		g/kWh		168.5		169.8				Density		kg/m³		995.4 @ 15 °C		Price		USD/t											
								Source of analysis																						
Cyl. lub.			REFERENCE		MEAS		Cyl. lub. system (Alpha):		REF		SET**		Cyl. lub. oil:		REF		ANALYS.													
	Absolute consumption		l/h		23.0		Target feed rate@MCR		g/kWh		0.60 0.80		BN		40 - 70		70													
	Spec. lub oil consumpt. *		g/kWh		0.88		Target f.r.@current load		g/kWh		0.80 SET based		Density		kg/m³		940.0 @ 15 °C													
	* Effective feed rate										** Acc.to OEM operation manuals and/or drainoil analysis				kg/m³		924 @ 40 °C													

Performance Report

Reading

Installation			Equipment				Type				No.							
APL DETROIT			MAIN ENGINE				10S90ME-C				DML0105360							
Cylinder condition	REFERENCE		MEASURED															
	REF.		AVG. ENGINE / CYL 1															
	Cool. water temp. CYL in	°C	75	76.0	76	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10				
	Cool. water temp. CYL out	°C	90	81.3	82	80	82	80	82	81	82	81	81	82				
	Cooling water press. CYL in	bar		4.9	4.9	4.9	4.9	4.9	4.9	4.9	4.9	4.9	4.9	4.9				
	Coolant temp. PISTON in	°C		43	43	43	43	43	43	43	43	43	43	43				
	Coolant temp. PISTON out	°C		54	53	53	54	53	55	53	54	54	53	53				
	Scav. air temp. PIST. UNDERS.	°C		47.3	47	44	49	49	50	45	48	47	46	49				
Crankcase			MEASURED															
			AVG. / ENG.	BRG 1	BRG 2	BRG 3	BRG 4	BRG 5	BRG 6	BRG 7	BRG 8	BRG 9	BRG 10	BRG 11	BRG 12	BRG 13		
	MAIN BEARING temp.	°C	51.4	52.7	52.2	48.9	51.5	50.5	51.2	51.5	53.6	53.3	49.0					
	Temp. deviation	°C		1	1	-3	0	-1	0	0	2	2	-2					
			CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10						
	CRANK PIN BEAR. temp	°C	49.5	48.6	50.1	52.6	48.9	50.1	48.4	50.2	48.7	48.9	48.2					
	Temp. deviation	°C		-1	1	3	-1	1	-1	1	-1	-1	-1					
	X-HEAD BEAR. temp	°C	48.5	47.3	47.8	47.8	49.1	49.7	48.9	48.1	48.3	49.3	48.9					
Temp. deviation	°C		-1	-1	-1	-1	1	1	0	0	0	1	0					
Liner wall			MEASURED															
			CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10						
	LINER WALL temp. aft	°C																
	LINER WALL temp. fp	°C		146	131	136	119	142	121	135	130	141	121					
	LINER WALL temp. fwd	°C																
	LINER WALL temp. exh	°C	AVG.	138	126	136	127	136	126	130	134	129	129					
LINER WALL temp. AVG	°C	132																
TC/SAC media			MEASURED															
			AVG.	TC 1	TC 2	TC 3												
	Water press. SAC in	bar	3.8	3.8	3.8	3.8												
	Lub oil temp. TC in	°C	42	42	42	42												
Lub oil temp. TC out	°C	60	60	60	60													
Engine media			MEAS								MEASURED							
	Lub oil temp ENGINE in	°C	43.0			Fuel oil press. ENGINE in	bar	7.50										
	Lub oil temp. THRUST SEGM.	°C	52.5			Fuel oil press. ENGINE out	bar	7.40										
	Lub oil press. SYSTEM OIL	bar	2.53			Fuel oil press. FILTER in	bar	7.6										
	Lub oil press. COOLING OIL	bar	2.46			Fuel oil press. FILTER out	bar	7.5										
	Lub oil press. TURB. OIL	bar	1.60			Hydraulic oil press. FILTER in	bar	2.6										
	Lub oil temp. TURB. OIL in	°C	41.80			Hydraulic oil press. FILTER out	bar	2.5										
	Lub oil temp. TURB. OIL out	°C	59.86			Hydraulic oil press. MAIN	bar	227.9										
	Axial vibration	mm	1.10			Sealing air press.	bar	7.4	PUMP 2	PUMP 3	PUMP 4	PUMP 5						
						Hydr. oil PUMP swash plate pos.	%	55.1	50.0	55.1								
Plan			MEAS															
	Exh. gas differential press. ECONOMIZER	mmWG																